

Wave 4.3 Mixture Density Heads Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved wave 4.3 dispersion-aware campaign: two- and three-component mixture-density heads for locally dispersed TE curves.

The campaign completed all planned entries

- 6 completed runs;
- 0 failed runs;
- 2 mixture profiles;
- 3 required direction surfaces: global , Fw , and Bw .

The runner-level scalar first entry is te_track2h_mdn_k2_bw . The program-level scalar winner did not change: te_periodic_gru_sequence_remote_Bw remains the best registry entry.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-06-13-11-11-47_track2h_mixture_density_heads_campaign_2026_06_13
Leaderboard	output/training_campaigns/2026-06-13-11-11-47_track2h_mixture_density_heads_campaign_2026_06_13/campaign_leaderboard.yaml
Best-run pointer	output/training_campaigns/2026-06-13-11-11-47_track2h_mixture_density_heads_campaign_2026_06_13/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-06-13-11-11-47_track2h_mixture_density_heads_campaign_2026_06_13/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/track_2/2026-06-13-10-40-25_track2h_mixture_density_heads_campaign_plan_report.md
Campaign package	config/training/track2h_mixture_density_heads/campaigns/2026-06-13_track2h_mixture_density_heads_campaign

Execution Summary

Field	Value
Campaign name	track2h_mixture_density_heads_campaign_2026_06_13
Completed runs	6
Failed runs	0
Model type	curve_aware_harmonic_residual_offset_probe
Profiles	mdn_k2 , mdn_k3
Tested surfaces	global , Fw , Bw
Runner-level scalar first entry	te_track2h_mdn_k2_bw
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Profile	Curve	Test MAE	Test RMSE	Val MAE
global	mdn_k2_global	mdn_k2	mixture expectation	0.003503	0.003938	0.003654
Fw	mdn_k3_fw	mdn_k3	mixture expectation	0.003235	0.003613	0.003253
Bw	mdn_k2_bw	mdn_k2	mixture expectation	0.002658	0.003198	0.002914

These rows are the branch-level closeout result. The Bw mdn_k2 row is also the campaign scalar leader, but it does not replace the global or Fw branch.

Mixture Density Leaderboard

Rank	Surface	Candidate	Profile	Test MAE	Test RMSE	Val MAE
1	Bw	mdn_k2_bw	mdn_k2	0.002658	0.003198	0.002914
2	Bw	mdn_k3_bw	mdn_k3	0.002721	0.003250	0.002775
3	Fw	mdn_k3_fw	mdn_k3	0.003235	0.003613	0.003253
4	Fw	mdn_k2_fw	mdn_k2	0.003339	0.003721	0.003285
5	global	mdn_k2_global	mdn_k2	0.003503	0.003938	0.003654
6	global	mdn_k3_global	mdn_k3	0.003564	0.003986	0.003617

Mixture Diagnostics Snapshot

Surface	Candidate	Entropy	Eff. Comp.	Mean Sigma	Separation
Bw	mdn_k2_bw	0.032	1.040	0.005300	0.012931
Bw	mdn_k3_bw	0.033	1.041	0.012381	0.036078
Fw	mdn_k3_fw	0.019	1.019	0.030782	0.048651
Fw	mdn_k2_fw	0.025	1.026	0.140972	0.039158
global	mdn_k2_global	0.016	1.016	0.135948	0.059065
global	mdn_k3_global	0.012	1.013	0.080886	0.107899

The mixture heads mostly collapsed toward one effective component. The scalar `Bw` gain is therefore useful, but it should be interpreted as useful MDN training pressure and expectation playback rather than confirmed learned multimodality.

Wave 4 series Dispersion-Aware Comparison

Surface	MDN	MDN MAE	Prob.	Prob. MAE	Robust	Robust MAE
global	mdn_k2_global	0.003503	gaussian_nll_global	0.003013	mae_robust_global	0.003406
Fw	mdn_k3_fw	0.003235	gaussian_nll_fw	0.003165	mae_robust_fw	0.003146
Bw	mdn_k2_bw	0.002658	quantile_p10_p50_p90_bw	0.002927	smooth_l1_robust_bw	0.003074

The MDN package is direction-selective. It improves the scalar `Bw` surface by 9.19% versus the previous best probabilistic `Bw` result and by about 13.5% versus the robust-loss `Bw` result. It does not improve the current `global` or `Fw` dispersion-aware scalar leaders.

Registry Effects

The campaign runner refreshed family registries for all six MDN families and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test MAE = 0.002344 .

Registry Scope	BestRun	Test MAE
Wave 4.3 global MDN	te_track2h_mdn_k2_global	0.003503
Wave 4.3 Fw MDN	te_track2h_mdn_k3_fw	0.003235
Wave 4.3 Bw MDN	te_track2h_mdn_k2_bw	0.002658
Current program scalar winner	te_periodic_gru_sequence_remote_Bw	0.002344

The best MDN branch remains about 13.41% worse than the current program scalar winner, so it is not a direct program promotion.

TE Curve Verification Pipeline Boundary

Official TE curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next TE Curve Verification Pipeline package should add all six mixture-density candidates and report accepted results separately for:

- `global` ;
- `Fw` ;
- `Bw` .

The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Wave 3.3, Wave 4.1 robust-loss, Wave 4.2 quantile/probabilistic, Wave 3.2, Wave 2.2, and the current accepted TE Curve Verification Pipeline baselines.

Closeout Decision

Wave 4.3 mixture-density execution is complete: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `mdn_k2_bw` forward as the strongest MDN scalar candidate and the strongest Wave 4 series scalar `Bw` candidate so far;
- do not carry MDN as the `global` or `Fw` default, because probabilistic and robust-loss candidates remain stronger there;
- treat the mixture-collapse diagnostics as a warning that this branch has not yet proven true multi-modal TE behavior;
- keep MDN alive until official TE curve verification decides whether the scalar `Bw` gain translates into curve-quality gain;
- do not promote Wave 4.3 MDN candidates before official TE curve verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched TE curve verification refresh for all six MDN candidates.
3. If the official TE Curve Verification refresh confirms the `Bw` gain, keep `mdn_k2_bw` available for later multi-head integration.
4. If the official TE Curve Verification refresh shows mixture collapse also at curve level, prioritize latent-state / hysteresis-aware models or the `wave 5.1` hybrid structured campaign before broadening MDN variants.
5. Keep `Fw`, `Bw`, and `global` as parallel branch decisions.